# Expense Tracker with Smart Insighter

## Test Case Document

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## 1. Testing Overview:

| Attribute | Details |
|---|---|
| Project Name | Expense Tracker Smart Insighter |
| Testing Type | Functional Testing, API Testing, UI Testing |
| Testing Tools | Swagger UI (FastAPI docs), Browser (Chrome), Postman, Manual Testing |
| Test Environment | Backend: localhost:8000 \| Frontend: localhost:5173 \| DB: PostgreSQL local |
| Total Test Cases | 21 |
| Passed | 21 |
| Failed | 0 |
| Skipped / N/A | 0 |

## 2. Authentication Test Cases:

| TC ID | Test Case | Input | Expected | Actual | Status |
|---|---|---|---|---|---|
| **TC-01** | Register new user | Valid name, email, password | User created, 200 OK | User created successfully | Pass |
| **TC-02** | Register - duplicate email | Email already in DB | 400: Email already registered | 400 returned correctly | Pass |
| **TC-03** | Login with valid credentials | Correct email + password | Login success | Login successful | Pass |
| **TC-04** | Login with wrong password | Correct email, wrong password | 401 Unauthorized | 401 returned correctly | Pass |
| **TC-05** | Login with unregistered email | Invalid email | 404 User not found | Error returned correctly | Pass |

## 3. Expense Management Test Cases:

| TC ID | Test Case | Input | Expected | Actual | Status |
|---|---|---|---|---|---|
| **TC-06** | Add valid expense | Amount + description | Expense added successfully | Expense stored correctly | Pass |
| **TC-07** | Add expense with empty amount | Empty amount field | Validation error | Validation error shown | Pass |
| **TC-08** | Add expense with invalid amount | Non-numeric value | Validation error | Error handled correctly | Pass |
| **TC-09** | View all expenses | None | List of expenses returned | Expenses displayed correctly | Pass |
| **TC-10** | View empty expense list | No data | Empty list message | Empty state handled | Pass |
| **TC-11** | Delete existing expense | Valid expense ID | Expense deleted | Deleted successfully | Pass |
| **TC-12** | Delete non-existing expense | Invalid ID | Error message | Error handled correctly | Pass |

## 4. Smart Categorization Test Cases:

| TC ID | Test Case | Input | Expected | Actual | Status |
|---|---|---|---|---|---|
| **TC-13** | Categorize food expense | “Pizza from Dominos” | Category = Food | Categorized correctly | Pass |
| **TC-14** | Categorize transport expense | “Uber ride” | Category = Transport | Categorized correctly | Pass |
| **TC-15** | Categorize shopping expense | “Bought clothes” | Category = Shopping | Categorized correctly | Pass |
| **TC-16** | Unknown keyword handling | “Random text” | Default category assigned | Default applied correctly | Pass |

## 5. API Test Cases:

| TC ID | API Endpoint | Method | Input | Expected Output | Status |
|---|---|---|---|---|---|
| **TC-17** | /register | POST | Valid user data | User created | Pass |
| **TC-18** | /login | POST | Valid credentials | Login success | Pass |
| **TC-19** | /expenses | POST | Expense data | Expense added | Pass |
| **TC-20** | /expenses | GET | None | Expense list returned | Pass |
| **TC-21** | /expenses/{id} | DELETE | Valid ID | Expense deleted | Pass |

## 6. Test Execution Summary:

| Metric | Value |
|---|---|
| **Total Test Cases Designed** | 21 |
| **Total Test Cases Executed** | 21 |
| **Pass Percentage** | 100% |
| **Fail Percentage** | 0% |
| **Test Completion Status** | Successful |